

Functional Requirements Document (FRD)

Algorithmic Trading System

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Introduction

The Algorithmic Trading System is a sophisticated platform designed to automate and optimize trading strategies across multiple asset classes, including stocks, Exchange-Traded Funds (ETFs), futures, options, forex, and cryptocurrencies. By integrating advanced artificial intelligence (AI) and machine learning technologies, the system enables traders, portfolio managers, and other

stakeholders to develop, backtest, and execute complex trading strategies efficiently and securely. This Functional Requirements Document (FRD) details the functional specifications required to build a robust, scalable, and user-friendly system that meets the diverse needs of its users.

The purpose of this FRD is to provide a clear and comprehensive blueprint for developers, architects, and stakeholders, ensuring alignment on the system's capabilities and facilitating development with AI-assisted coding tools such as Kilo Code. The document outlines all functional aspects, from user management to advanced trading features, and serves as a foundation for subsequent technical design and implementation phases.

Scope

The Algorithmic Trading System encompasses a wide range of functionalities to support automated trading and strategy optimization. The scope includes:

- **User Management:** Secure registration, authentication, and role-based access control.
- **Core Trading Infrastructure:** Multi-asset trading with broker integrations and order management.
- **Data Management and Analytics:** Real-time and historical data handling with advanced analytics tools.
- **AI-Powered Strategy Development:** Tools for creating and backtesting trading strategies using AI and machine learning.
- **Agentic AI Assistant:** Natural language interaction for trading commands and document analysis.
- **User Interface:** Responsive web and mobile access with personalized dashboards and charting.
- **Security and Compliance:** Enterprise-grade features for regulatory adherence and risk management.
- **Options Trading:** Specialized tools for options strategies and risk analysis.

Out of Scope:

- Development or modification of third-party broker systems beyond integration points.

- Management of external data sources beyond specified integrations and fallback mechanisms.
- Hardware infrastructure provisioning, assumed to be managed by deployment teams.

This scope ensures a focused development effort on core trading functionalities while leveraging external systems where appropriate.

Functional Requirements

The following sections detail the specific functional capabilities the Algorithmic Trading System must provide, organized by major components.

User Management

- **FR-UM-01:** The system shall allow users to register with a unique email address and password.
- **FR-UM-02:** The system shall enforce strong password policies (minimum 12 characters, including uppercase, lowercase, numbers, and special characters).
- **FR-UM-03:** The system shall support multi-factor authentication (MFA) using time-based one-time passwords (TOTP) or authenticator apps.
- **FR-UM-04:** The system shall enable users to update their profile information (e.g., name, contact details, preferences).
- **FR-UM-05:** The system shall integrate Single Sign-On (SSO) with major identity providers (e.g., Google, Okta).
- **FR-UM-06:** The system shall provide a secure password reset mechanism with email verification and temporary links (expiring within 24 hours).
- **FR-UM-07:** The system shall log all user authentication events for audit purposes.

Core Trading Infrastructure

- **FR-CTI-01:** The system shall integrate with multiple brokers (e.g., Interactive Brokers, Alpaca, TD Ameritrade) via their respective APIs.
- **FR-CTI-02:** The system shall support trading across multiple asset classes: stocks, ETFs, futures, options, forex, and cryptocurrencies.

- **FR-CTI-03:** The system shall include an Order Management System (OMS) supporting:
 - Basic order types: Market, Limit, Stop, Stop-Limit, Trailing Stop.
 - Advanced order types: Good-Til-Canceled (GTC), Immediate-or-Cancel (IOC), Fill-or-Kill (FOK), Volume-Weighted Average Price (VWAP), Time-Weighted Average Price (TWAP), Iceberg, One-Cancels-the-Other (OCO).
- **FR-CTI-04:** The system shall provide real-time order status updates (e.g., pending, executed, canceled) with user notifications via email or in-app alerts.
- **FR-CTI-05:** The system shall log all orders and executions with timestamps and error details for failed transactions.
- **FR-CTI-06:** The system shall support seamless switching between paper trading (simulated) and live trading modes.
- **FR-CTI-07:** The system shall validate orders against user-defined risk limits before submission.

Data Management and Analytics

- **FR-DMA-01:** The system shall integrate with free market data sources (e.g., Yahoo Finance, Alpha Vantage) for paper trading.
- **FR-DMA-02:** The system shall integrate with Interactive Brokers for live trading data feeds.
- **FR-DMA-03:** The system shall implement a fallback mechanism to switch data sources in case of primary source failure.
- **FR-DMA-04:** The system shall store historical market data with a minimum depth of 5 years, accessible via query.
- **FR-DMA-05:** The system shall integrate alternative data sources, such as real-time news feeds with sentiment analysis.
- **FR-DMA-06:** The system shall provide advanced analytics tools, including:
 - Market screener with natural language filter support (e.g., "stocks with P/E < 15 and volume > 1M").
 - AI-powered opportunity detection (e.g., anomaly detection, trend analysis).
 - Market microstructure analysis (e.g., order book depth, liquidity metrics).

- **FR-DMA-07:** The system shall ensure data integrity with checksum validation and error logging.

AI-Powered Strategy Development and Backtesting

- **FR-AIS-01:** The system shall provide a no-code strategy builder with a drag-and-drop interface for creating trading rules.
- **FR-AIS-02:** The system shall include a generative AI tool to convert natural language prompts (e.g., "buy when RSI < 30") into executable strategy code.
- **FR-AIS-03:** The system shall offer a Python-based studio with cloud-hosted Jupyter notebooks for advanced users.
- **FR-AIS-04:** The system shall provide an enterprise-grade backtesting engine supporting:
 - Realistic modeling (e.g., slippage, commissions, market impact).
 - Advanced analytics (e.g., Monte Carlo stress testing, walk-forward optimization).
 - Multi-asset and multi-timeframe simulations.
- **FR-AIS-05:** The system shall include a strategy marketplace for users to share, license, or purchase trading strategies.
- **FR-AIS-06:** The system shall support version control for strategies using Git integration with commit history.
- **FR-AIS-07:** The system shall generate detailed backtest reports with performance metrics (e.g., Sharpe Ratio, max drawdown).

Agentic AI Assistant

- **FR-AIA-01:** The system shall provide an AI Assistant capable of processing natural language queries (e.g., "What's the current price of AAPL?").
- **FR-AIA-02:** The AI Assistant shall execute trading commands based on user instructions (e.g., "Buy 100 shares of TSLA at market").
- **FR-AIA-03:** The AI Assistant shall analyze user-uploaded documents (e.g., PDFs, CSVs) and answer questions using Agentic Retrieval-Augmented Generation (RAG).
- **FR-AIA-04:** The system shall implement a collaborative agent network for task delegation (e.g., data retrieval, strategy execution).
- **FR-AIA-05:** The AI Assistant shall improve over time through continuous learning from user interactions.

- **FR-AIA-06:** The system shall provide tools for custom fine-tuning of AI models with user-uploaded datasets.
- **FR-AIA-07:** The AI Assistant shall support multi-modal inputs (text, voice) for accessibility.

User Experience and Multi-Platform Accessibility

- **FR-UE-01:** The system shall provide a unified, responsive web interface built with Next.js, featuring personalized dashboards.
- **FR-UE-02:** The interface shall integrate institutional-grade charting tools (e.g., TradingView) with customizable indicators.
- **FR-UE-03:** The system shall include a conversational chatbot interface for the AI Assistant, embedded in the dashboard.
- **FR-UE-04:** The system shall support multi-platform access:
 - Web application with real-time updates.
 - Progressive Web App (PWA) with offline capabilities for basic functions.
- **FR-UE-05:** The system shall comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards.
- **FR-UE-06:** The system shall support multi-language interfaces with real-time translation (e.g., English, Spanish, Mandarin).
- **FR-UE-07:** The system shall allow users to customize dashboard widgets (e.g., watchlists, performance charts).

Enterprise-Grade Security, Compliance, and Operations

- **FR-SEC-01:** The system shall implement a Zero-Trust Architecture with continuous identity verification.
- **FR-SEC-02:** The system shall comply with regulatory standards, including GDPR, MiFID II, and SEC regulations.
- **FR-SEC-03:** The system shall provide automated compliance reporting and immutable audit trails with timestamps.
- **FR-SEC-04:** The system shall include a real-time risk management hub with pre-trade checks (e.g., position limits, margin requirements).
- **FR-SEC-05:** The system shall support operational monitoring with Prometheus and Grafana dashboards.
- **FR-SEC-06:** The system shall enforce role-based access control (RBAC) with customizable roles and permissions.

- **FR-SEC-07:** The system shall encrypt all sensitive data at rest (AES-256) and in transit (TLS 1.3).

Options Trading

- **FR-OPT-01:** The system shall support trading of options strategies (e.g., straddles, spreads, iron condors).
- **FR-OPT-02:** The system shall provide tools for calculating options Greeks (Delta, Gamma, Theta, Vega, Rho) and generating payoff diagrams.
- **FR-OPT-03:** The system shall include a volatility suite with surface constructors and historical volatility analysis.
- **FR-OPT-04:** The system shall offer risk management tools specific to options (e.g., margin calculators, stress testing).
- **FR-OPT-05:** The system shall provide real-time implied volatility surfaces for listed options.
- **FR-OPT-06:** The system shall integrate dividend forecasts into options pricing models.
- **FR-OPT-07:** The system shall support exotic options (e.g., binary, barrier) with customizable parameters.

User Roles and Permissions

The system shall define the following user roles with associated permissions:

Role	Permissions
Admin	Full system access, user management, system configuration, audit log access
Trader	Place and manage orders, view market data and analytics, access personal dashboard
Quant	Develop, backtest, and deploy trading strategies, access Python studio
Risk Manager	Set risk parameters, monitor positions and compliance, approve high-risk trades

Role	Permissions
Compliance Officer	Access audit trails, generate compliance reports, enforce regulatory policies

Permissions shall be enforced through RBAC, ensuring granular control over system access and actions.

System Interfaces

The system shall interact with the following external systems and interfaces:

- **Broker APIs:** Trade execution and account management (e.g., Interactive Brokers REST API, Alpaca WebSocket).
- **Market Data Feeds:** Real-time and historical data (e.g., Yahoo Finance API, Interactive Brokers data feed).
- **AI Models:** Local Large Language Models (LLMs) via Ollama for secure, on-premises processing.
- **User Interfaces:** Web and mobile access via HTTPS with responsive design.
- **Document Processing:** Unstructured.io for parsing uploaded files (e.g., PDFs, Word documents, CSVs).

All interfaces shall use standardized protocols (e.g., REST, WebSocket) and include error handling for connectivity issues.

Data Requirements

The system shall manage the following data types:

- **User Data:** Profiles (name, email, preferences), authentication credentials, role assignments.
- **Market Data:** Real-time quotes, historical prices, volume, and order book data for supported asset classes.
- **Strategy Data:** Code (Python, no-code rules), parameters, backtest results, and version history.
- **Order and Trade Logs:** Order details (type, quantity, price), execution timestamps, and status updates.

- **AI Model Data:** Inputs (queries, documents), outputs (responses, strategies), and embeddings for RAG.

Data shall be stored in a secure, scalable database with indexing for efficient retrieval and backup mechanisms for disaster recovery.

Non-Functional Requirements

While detailed in a separate document, key non-functional requirements include:

- **Performance:** Order execution latency < 100ms under normal conditions.
- **Scalability:** Horizontal scaling to support 10,000 concurrent users and 1M daily trades.
- **Security:** Compliance with OWASP Top 10 and financial industry standards (e.g., PCI DSS).
- **Reliability:** 99.99% uptime with automated failover and disaster recovery procedures.
- **Usability:** Intuitive interface with a learning curve of < 1 hour for basic functions.

Assumptions and Constraints

Assumptions:

- Users have basic knowledge of trading concepts and terminology.
- Market data and broker APIs are available and reliable within specified tolerances.
- Sufficient computational resources are available for AI model training and backtesting.

Constraints:

- Must leverage specified open-source technologies (e.g., Next.js, Ollama, Prometheus).
- Must adhere to a 5-phase development plan as outlined in the project roadmap.

- Must comply with all applicable financial regulations across supported jurisdictions.

Appendices

Glossary

- **Agentic RAG:** Retrieval-Augmented Generation with AI agents for enhanced query processing.
- **OMS:** Order Management System for handling trade orders and executions.
- **Greeks:** Risk metrics for options (e.g., Delta, Gamma, Theta, Vega, Rho).

References

- Product Requirements Document (PRD)
- Technical Specification Document (TSD)
- API Documentation for Integrated Brokers and Data Sources
- Comprehensive Test Plan
- Features, Phases & Integration Strategy

This Functional Requirements Document (FRD) provides a detailed and actionable specification for the Algorithmic Trading System, ensuring all functional aspects are clearly defined for development, testing, and deployment.